

1. Grammar, sentence completion and vocabulary — Concept foundation

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The central idea is to build a chain from representation to operation to consequence. In **Grammar, sentence completion and vocabulary**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Grammar, sentence completion and vocabulary can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Grammar, sentence completion and vocabulary. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

GATE traps. Common mistakes around Grammar, sentence completion and vocabulary include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Grammar, sentence completion and vocabulary, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

1. Grammar, sentence completion and vocabulary — Deep dive

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Grammar, sentence completion and vocabulary with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Grammar, sentence completion and vocabulary, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Grammar, sentence completion and vocabulary, a 20–40 second classification step often saves several minutes of algebra.

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Grammar, sentence completion and vocabulary. Edge cases are where apparently correct reasoning often breaks.

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Formula sheet. Relevant formulas and relationships include: Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $nPr = n! / (n-r)!$; $nCr = n! / (r!(n-r)!)$; Average = sum / n . Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

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The central idea is to build a chain from representation to operation to consequence. In **Reading comprehension and inference**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Reading comprehension and inference, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Reading comprehension and inference. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

GATE traps. Common mistakes around Reading comprehension and inference include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Reading comprehension and inference, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

2. Reading comprehension and inference — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Reading comprehension and inference, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Reading comprehension and inference. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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2. Reading comprehension and inference — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Reading comprehension and inference. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Reading comprehension and inference. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$nPr = n! / (n-r)!$$

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2. Reading comprehension and inference — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $nPr = n! / (n-r)!$; $nCr = n! / (r!(n-r)!)$; Average = sum / n . Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Reading comprehension and inference. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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2. Reading comprehension and inference — Unit synthesis

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Examiner perspective. A strong question on Reading comprehension and inference can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Reading comprehension and inference. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Average = sum/n

GATE traps. Common mistakes around Reading comprehension and inference include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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3. Quantitative arithmetic: ratios, percentages and averages — Concept foundation

For **Quantitative arithmetic: ratios, percentages and averages**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Worked reasoning. Consider a small exam-sized instance related to Quantitative arithmetic: ratios, percentages and averages. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

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3. Quantitative arithmetic: ratios, percentages and averages — Deep dive

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$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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3. Quantitative arithmetic: ratios, percentages and averages — Worked example

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3. Quantitative arithmetic: ratios, percentages and averages — GATE traps

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3. Quantitative arithmetic: ratios, percentages and averages — Comparison & intuition

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Quantitative arithmetic: ratios, percentages and averages with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Quantitative arithmetic: ratios, percentages and averages. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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3. Quantitative arithmetic: ratios, percentages and averages — Derivation / proof idea

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3. Quantitative arithmetic: ratios, percentages and averages — Edge cases

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3. Quantitative arithmetic: ratios, percentages and averages — Formula & memory map

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Worked reasoning. Consider a small exam-sized instance related to Quantitative arithmetic: ratios, percentages and averages. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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Average = sum/n

GATE traps. Common mistakes around Quantitative arithmetic: ratios, percentages and averages include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Quantitative arithmetic: ratios, percentages and averages, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

4. Profit, loss, mixtures and simple interest — Concept foundation

For **Profit, loss, mixtures and simple interest**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Profit, loss, mixtures and simple interest**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Profit, loss, mixtures and simple interest can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Profit, loss, mixtures and simple interest. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

GATE traps. Common mistakes around Profit, loss, mixtures and simple interest include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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4. Profit, loss, mixtures and simple interest — Deep dive

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High-yield formula / relation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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4. Profit, loss, mixtures and simple interest — Worked example

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High-yield formula / relation

$$nPr = n! / (n-r)!$$

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High-yield formula / relation

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4. Profit, loss, mixtures and simple interest — Comparison & intuition

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Profit, loss, mixtures and simple interest with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Profit, loss, mixtures and simple interest. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation
Average=sum/n

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4. Profit, loss, mixtures and simple interest — Derivation / proof idea

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Profit, loss, mixtures and simple interest, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Profit, loss, mixtures and simple interest. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

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4. Profit, loss, mixtures and simple interest — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Profit, loss, mixtures and simple interest, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Profit, loss, mixtures and simple interest. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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4. Profit, loss, mixtures and simple interest — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Profit, loss, mixtures and simple interest. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Profit, loss, mixtures and simple interest. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$nPr = \frac{n!}{(n-r)!}$$

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4. Profit, loss, mixtures and simple interest — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $nPr = n! / (n-r)!$; $nCr = n! / (r!(n-r)!)$; Average = sum / n . Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Profit, loss, mixtures and simple interest. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$nCr = n! / (r!(n-r)!)$$

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4. Profit, loss, mixtures and simple interest — Unit synthesis

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Examiner perspective. A strong question on Profit, loss, mixtures and simple interest can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Profit, loss, mixtures and simple interest. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Average = sum/n

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5. Permutations, combinations and probability — Concept foundation

For **Permutations, combinations and probability**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Worked reasoning. Consider a small exam-sized instance related to Permutations, combinations and probability. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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5. Permutations, combinations and probability — Deep dive

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High-yield formula / relation

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5. Permutations, combinations and probability — Worked example

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Permutations, combinations and probability with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Permutations, combinations and probability. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Average = sum/n

GATE traps. Common mistakes around Permutations, combinations and probability include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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5. Permutations, combinations and probability — Derivation / proof idea

For **Permutations, combinations and probability**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Permutations, combinations and probability, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Permutations, combinations and probability. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

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5. Permutations, combinations and probability — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Permutations, combinations and probability, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Permutations, combinations and probability. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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5. Permutations, combinations and probability — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Permutations, combinations and probability. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Permutations, combinations and probability. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$nPr = n! / (n-r)!$$

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5. Permutations, combinations and probability — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $nPr = n! / (n-r)!$; $nCr = n! / (r!(n-r)!)$; Average = sum / n . Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Permutations, combinations and probability. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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5. Permutations, combinations and probability — Unit synthesis

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Examiner perspective. A strong question on Permutations, combinations and probability can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Permutations, combinations and probability. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Average = sum/n

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6. Number systems, logarithms and sequences — Concept foundation

For **Number systems, logarithms and sequences**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

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6. Number systems, logarithms and sequences — Deep dive

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High-yield formula / relation

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6. Number systems, logarithms and sequences — Worked example

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6. Number systems, logarithms and sequences — GATE traps

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6. Number systems, logarithms and sequences — Comparison & intuition

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Number systems, logarithms and sequences with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

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6. Number systems, logarithms and sequences — Edge cases

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Topic summary. By the end of this section you should be able to define the core terms of Number systems, logarithms and sequences, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

6. Number systems, logarithms and sequences — Unit synthesis

For **Number systems, logarithms and sequences**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Number systems, logarithms and sequences**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Number systems, logarithms and sequences can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Number systems, logarithms and sequences. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Average = sum/n

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7. Geometry, mensuration and data interpretation — Concept foundation

For **Geometry, mensuration and data interpretation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

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7. Geometry, mensuration and data interpretation — Deep dive

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High-yield formula / relation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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7. Geometry, mensuration and data interpretation — Worked example

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High-yield formula / relation

$$nPr = \frac{n!}{(n-r)!}$$

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7. Geometry, mensuration and data interpretation — GATE traps

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High-yield formula / relation

$$nC_r = \frac{n!}{r!(n-r)!}$$

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7. Geometry, mensuration and data interpretation — Comparison & intuition

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Geometry, mensuration and data interpretation with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Geometry, mensuration and data interpretation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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7. Geometry, mensuration and data interpretation — Derivation / proof idea

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Geometry, mensuration and data interpretation, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Geometry, mensuration and data interpretation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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7. Geometry, mensuration and data interpretation — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Geometry, mensuration and data interpretation, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Geometry, mensuration and data interpretation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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7. Geometry, mensuration and data interpretation — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Geometry, mensuration and data interpretation. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Geometry, mensuration and data interpretation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$nPr = \frac{n!}{(n-r)!}$$

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7. Geometry, mensuration and data interpretation — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $nPr = n! / (n-r)!$; $nCr = n! / (r!(n-r)!)$; Average = sum / n . Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Geometry, mensuration and data interpretation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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Worked reasoning. Consider a small exam-sized instance related to Geometry, mensuration and data interpretation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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8. Analytical reasoning, arrangements and deduction — Concept foundation

For **Analytical reasoning, arrangements and deduction**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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8. Analytical reasoning, arrangements and deduction — Deep dive

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High-yield formula / relation

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8. Analytical reasoning, arrangements and deduction — Worked example

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Worked reasoning. Consider a small exam-sized instance related to Analytical reasoning, arrangements and deduction. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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High-yield formula / relation

$$nC_r = n! / (r!(n-r)!)$$

GATE traps. Common mistakes around Analytical reasoning, arrangements and deduction include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Analytical reasoning, arrangements and deduction, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

8. Analytical reasoning, arrangements and deduction — Comparison & intuition

For **Analytical reasoning, arrangements and deduction**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Analytical reasoning, arrangements and deduction**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Analytical reasoning, arrangements and deduction with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Analytical reasoning, arrangements and deduction. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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8. Analytical reasoning, arrangements and deduction — Derivation / proof idea

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Analytical reasoning, arrangements and deduction, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Analytical reasoning, arrangements and deduction. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

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8. Analytical reasoning, arrangements and deduction — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Analytical reasoning, arrangements and deduction, a 20–40 second classification step often saves several minutes of algebra.

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8. Analytical reasoning, arrangements and deduction — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Analytical reasoning, arrangements and deduction. Edge cases are where apparently correct reasoning often breaks.

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High-yield formula / relation

$$nPr = \frac{n!}{(n-r)!}$$

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8. Analytical reasoning, arrangements and deduction — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $nPr = n! / (n-r)!$; $nCr = n! / (r!(n-r)!)$; Average = sum / n . Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Analytical reasoning, arrangements and deduction. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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8. Analytical reasoning, arrangements and deduction — Unit synthesis

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Examiner perspective. A strong question on Analytical reasoning, arrangements and deduction can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Analytical reasoning, arrangements and deduction. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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Average = sum/n

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9. Patterns, spatial reasoning and numerical logic — Concept foundation

For **Patterns, spatial reasoning and numerical logic**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation

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9. Patterns, spatial reasoning and numerical logic — Deep dive

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9. Patterns, spatial reasoning and numerical logic — Worked example

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High-yield formula / relation

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9. Patterns, spatial reasoning and numerical logic — Formula & memory map

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The central idea is to build a chain from representation to operation to consequence. In **Patterns, spatial reasoning and numerical logic**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Formula sheet. Relevant formulas and relationships include: Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $nPr = n! / (n-r)!$; $nCr = n! / (r!(n-r)!)$; Average = sum / n . Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Patterns, spatial reasoning and numerical logic. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$nCr = n! / (r!(n-r)!)$$

GATE traps. Common mistakes around Patterns, spatial reasoning and numerical logic include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Patterns, spatial reasoning and numerical logic, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

9. Patterns, spatial reasoning and numerical logic — Unit synthesis

For **Patterns, spatial reasoning and numerical logic**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Examiner perspective. A strong question on Patterns, spatial reasoning and numerical logic can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Patterns, spatial reasoning and numerical logic. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Average = sum/n

GATE traps. Common mistakes around Patterns, spatial reasoning and numerical logic include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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10. GATE GA mixed practice and time-management strategy — Concept foundation

For **GATE GA mixed practice and time-management strategy**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Examiner perspective. A strong question on GATE GA mixed practice and time-management strategy can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$

GATE traps. Common mistakes around GATE GA mixed practice and time-management strategy include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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10. GATE GA mixed practice and time-management strategy — Deep dive

For **GATE GA mixed practice and time-management strategy**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Examiner perspective. A strong question on GATE GA mixed practice and time-management strategy can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

GATE traps. Common mistakes around GATE GA mixed practice and time-management strategy include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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10. GATE GA mixed practice and time-management strategy — Worked example

For **GATE GA mixed practice and time-management strategy**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

$$nPr = \frac{n!}{(n-r)!}$$

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10. GATE GA mixed practice and time-management strategy — GATE traps

For **GATE GA mixed practice and time-management strategy**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation

$$nC_r = n! / (r!(n-r)!)$$

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10. GATE GA mixed practice and time-management strategy — Comparison & intuition

For **GATE GA mixed practice and time-management strategy**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for GATE GA mixed practice and time-management strategy with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Average = sum/n

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10. GATE GA mixed practice and time-management strategy — Derivation / proof idea

For **GATE GA mixed practice and time-management strategy**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For GATE GA mixed practice and time-management strategy, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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10. GATE GA mixed practice and time-management strategy — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For GATE GA mixed practice and time-management strategy, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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10. GATE GA mixed practice and time-management strategy — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for GATE GA mixed practice and time-management strategy. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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10. GATE GA mixed practice and time-management strategy — Formula & memory map

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Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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10. GATE GA mixed practice and time-management strategy — Unit synthesis

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Examiner perspective. A strong question on GATE GA mixed practice and time-management strategy can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to GATE GA mixed practice and time-management strategy. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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